

Part 2: Arithmetic, Inputs, and Variables in Python

1 Arithmetic Operators in Python

Python arithmetic operators perform mathematical operations on numbers.

Operator	Purpose	Example	Result
+	Addition	2 + 4	6
-	Subtraction	10 - 2.5	7.5
*	Multiplication	-2 * 123	-246
/	Division (float result)	9 / 2	4.5
//	Division (integer result)	9 // 2	4
%	Modulo (remainder)	9 % 2	1
**	Exponentiation	2 ** 3	8

Order of operations:

Exponents → Multiplication/Division → Addition/Subtraction

Parentheses () can override the order.

```
print(2 + 3 * 3)
print((2 + 3) * 3)
```

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[title=Concept Summary]

- Operators act on operands to perform calculations.
- Parentheses change the order of operations.
- // performs integer (floor) division.

2 Operands, Operators, and Data Types

A calculation consists of operands and operators. The type of operands determines the result type:

- Integer + Integer → Integer
- Float + Float → Float
- Integer + Float → Float

Exception: Division / always returns a float.

$$\frac{1}{5} = 0.2$$

```
height = 172.5
weight = 68.55
```

```
# Body Mass Index (BMI)
bmi = weight / (height / 100) ** 2

print(f"The BMI is {bmi}")
```

The BMI is 23.037177063642087

$$\text{BMI} = \frac{\text{weight (kg)}}{(\text{height (m)})^2}$$

[title=Concept Summary]

- Result type depends on operand types.
- Division / always produces a float.
- Integer division // rounds down to nearest integer.

3 Numbers as Input

Integer Input

```
year = int(input("Which year were you born? "))
print(f"Your age at the end of the year 2021: {2021 - year}")
```

Which year were you born? 1995 Your age at the end of the year 2021: 26

Floating Point Input

```
height = float(input("What is your height? "))
weight = float(input("What is your weight? "))
```

```
height = height / 100
bmi = weight / height ** 2
```

```
print(f"The BMI is {bmi}")
```

What is your height? 163 What is your weight? 74.45 The BMI is 28.02137829801649

[title=Concept Summary]

- input() returns a string.
- Use int() or float() to convert for numeric calculations.

4 Using Variables

Sum of Three Numbers

```
number1 = int(input("First number: "))
number2 = int(input("Second number: "))
number3 = int(input("Third number: "))

sum = number1 + number2 + number3
print(f"The sum of the numbers: {sum}")
```

First number: 5 Second number: 21 Third number: 7 The sum of the numbers: 33

Reusing Variables with +=

```
sum = 0

sum += int(input("First number: "))
sum += int(input("Second number: "))
sum += int(input("Third number: "))

print(f"The sum of the numbers: {sum}")
```

[title=Concept Summary]

- The += operator adds to the variable and stores the result.
- Reuse variables only for similar types of input.

Descriptive Variable Names

```
name = input("What is your name? ")
print("Hi " + name + "!")

age = int(input("What is your age? "))
```

[title=Concept Summary]

- Use descriptive variable names for clarity.
- Avoid reusing variables for unrelated data.

5 Comparison Operators (Optional Table Example)

Operator	Purpose	Example	Result
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	7 > 4	True
<	Less than	7 < 4	False
>=	Greater than or equal to	8 >= 8	True
<=	Less than or equal to	6 <= 3	False